

b) Regraph $f(x)$ on the interval $.70 \leq x \leq .71$ to get two decimal places of accuracy in the location of the x -intercept. Continue this process until you have 7 places of accuracy. What is the x -intercept?

[Answer: $x = .7071067812\dots$]

The circular functions Graphing packages “know” the familiar functions of trigonometry. Trigonometric functions are qualitatively different from the functions in the preceding problems. Those functions are defined by algebraic formulas, so they are called **algebraic functions**.

The trigonometric functions are defined by explicit “recipes,” but *not* by algebraic formulas; they are called **transcendental functions**. For calculus, we always use the definition of the trigonometric functions as **circular functions**. This definition begins with a unit circle centered at the origin. Given the input number t , locate a point P on the circle by tracing an arc of length t along the circle from the point $(1, 0)$. If t is positive, trace the arc counterclockwise; if t is negative, trace it clockwise. Because the circle has radius 1, the arc of length t subtends a central angle of **radian** measure t .

The circular (or trigonometric) functions $\cos t$ and $\sin t$ are defined as the coordinates of the point P ,

$$P = (\cos t, \sin t).$$

The other trigonometric functions are defined in terms of the sine and cosine:

$$\begin{aligned}\tan t &= \sin t / \cos t & \sec t &= 1 / \cos t \\ \cot t &= \cos t / \sin t & \csc t &= 1 / \sin t\end{aligned}$$

Notice that when t is a positive acute angle, the circle definition agrees with the right triangle definitions of the sine and cosine:

$$\sin t = \frac{\text{opposite}}{\text{hypotenuse}} \quad \text{and} \quad \cos t = \frac{\text{adjacent}}{\text{hypotenuse}}.$$

However, the circle definitions of the sine and cosine have the important advantage that they produce functions whose domains are the set of *all* real numbers. (What are the domains of the tangent, secant and cosecant functions?) In calculus, not only are the circle definitions

